

Mohamed Wael Mohamed

AI & Data Analytics | Applied Machine Learning | Agricultural Engineer

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PROFESSIONAL SUMMARY

Agricultural Engineering graduate (Irrigation, Mansoura University 2023, GPA 2.9/4.0) enrolled in Egypt's Ministry of Communications & IT Digital Pioneers Initiative (Applied AI & Data Analytics, Dec 2025 – Sep 2026). Built two production-grade machine learning web applications end-to-end. Holds four professional certifications in data analytics, cloud computing, and Python. Strong foundation in statistics, data mining, deep learning, and full-stack ML deployment.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript, C (Arduino / Embedded)

Machine Learning & AI: Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch (basics), Random Forest, SVM, KNN, Logistic Regression, Neural Networks (MLP, CNN)

Data Analysis & Mining: Pandas, NumPy, SciPy, Applied Statistics, Feature Engineering, EDA, Data Visualization, Matplotlib, Seaborn

Web & API Development: FastAPI, REST API design, React, Next.js 14, Vite, Framer Motion

Cloud & DevOps: Microsoft Azure (AZ-900), Google App Engine, Git, GitHub

Deep Learning: TensorFlow, Keras, PyTorch (basics), CNNs, RNNs, Transfer Learning, Model Training & Evaluation

Business Intelligence: Power BI, Tableau, Microsoft Excel, Interactive Dashboards

Hardware / IoT: Arduino, Embedded C, IoT Sensors, Smart Systems

PROJECTS

Hotel Intelligence Dashboard – Cancellation Prediction System

2025

Stack: Python · FastAPI · Scikit-learn · React · Vite · Google App Engine · Claude AI

- Developed an end-to-end ML web application predicting hotel booking cancellations using Logistic Regression trained on 119,390 records (37% baseline cancellation rate); achieved 82% accuracy after feature engineering.
- Engineered 24 features from raw booking data: lead-time segmentation, ADR (Average Daily Rate) normalisation, seasonal demand flags, and repeat-guest indicators.
- Designed React + Vite SPA with live KPI cards, interactive prediction form, 5-phase workflow visualisation, and Claude AI chatbot integration.
- Deployed production FastAPI backend (endpoints: /predict, /stats, /chatbot) on Google App Engine with app.yaml CI configuration.

- Keywords: classification, logistic regression, feature engineering, REST API, deployment, data pipeline, hotel analytics.

Body Performance Intelligence System – 6-Model Classifier

2025

Stack: Python · FastAPI · XGBoost · Scikit-learn · Next.js 14 · Framer Motion · Claude AI

- Classified athletic performance into four tiers (A–D) using six ML models: Random Forest 75.4% (best), Neural Network 73.8%, XGBoost 71.8%, SVM 70.0%, Decision Tree 69.2%, Logistic Regression 63.8%.
- Performed comprehensive feature engineering: created 10 composite features (Fitness Score $r=+0.47$, Body Fat Category $r=-0.38$, Grip/Weight Ratio, BMI, MAP, Pulse Pressure, Relative Strength, Age Group encoding).
- Built full EDA module with 7 statistical visualisations (histograms, boxplots, scatter plots, correlation heatmap, confusion matrix) with ANOVA significance annotations.
- Implemented personalised Nutrition Plan (calories, macros, food lists) and weekly Training Schedule auto-generated per predicted performance class.
- Keywords: multi-class classification, XGBoost, random forest, feature engineering, EDA, data visualisation, full-stack ML, Next.js.

Smart Irrigation System – Graduation Project

2023

Stack: Arduino · Embedded C · Soil-moisture / Temperature / Humidity Sensors | Grade: Excellent

- Designed and implemented an automated irrigation controller using Arduino microcontrollers and environmental sensors to optimise water delivery based on real-time field data.
- Programmed decision logic in Embedded C to maintain optimal soil-moisture levels, reducing water consumption compared to fixed-schedule methods.
- Awarded Excellent grade (highest available) from the Faculty of Agriculture, Mansoura University.

WORK EXPERIENCE

Sales Representative – Computers & Laptops

2023 – 2025

Technology Retail Sector, Mansoura, Egypt

- Provided technical consultations for hardware and software purchases, consistently meeting monthly sales targets.
- Built long-term client relationships; developed strong communication, negotiation, and problem-solving skills through daily client interactions.

AI Trainee – Applied AI & Data Analytics

Dec 2025 – Sep 2026 (ongoing)

Digital Pioneers Initiative – Ministry of Communications & IT, Egypt | 9-month intensive programme

- Selected among thousands of applicants for a highly competitive 9-month government-sponsored AI programme run by Egypt's Ministry of Communications & IT, targeting top graduates in technology and engineering.
- Completed hands-on training in Applied Statistics, Data Mining & Visualisation, Machine Learning, Deep Learning, AI Tools, Power BI, and Microsoft Azure.
- Applied acquired skills by building two end-to-end ML projects: a Hotel Cancellation Prediction System (119,390 records, Logistic Regression) and a Body Performance Classification System (6 ML models, XGBoost, Random Forest 75.4% accuracy).
- Developed expertise in full-stack ML deployment (FastAPI, Google App Engine), feature engineering, EDA, and data visualisation using Python, Pandas, Scikit-learn, and Power BI.

- Collaborated in a team environment on data-driven projects, practising agile workflows, code review, and technical presentation of analytical findings to peers and instructors.

EDUCATION

B.Sc. Agricultural Engineering – Irrigation Specialisation

Sep 2019 – Jun 2023

Faculty of Agriculture, Mansoura University, Egypt

- Cumulative GPA: Good with High Distinction (2.9 / 4.0)
- Graduation project: Smart Irrigation System using Arduino – Grade: Excellent

CERTIFICATIONS

- Microsoft Power BI Certification
- Microsoft Azure Fundamentals – AZ-900
- Python Programming – DataCamp
- Google Data Analytics Professional Certificate – Google / Coursera

LANGUAGES

- Arabic – Native
- English – Upper-Intermediate (B2)
- German – Beginner (A1)